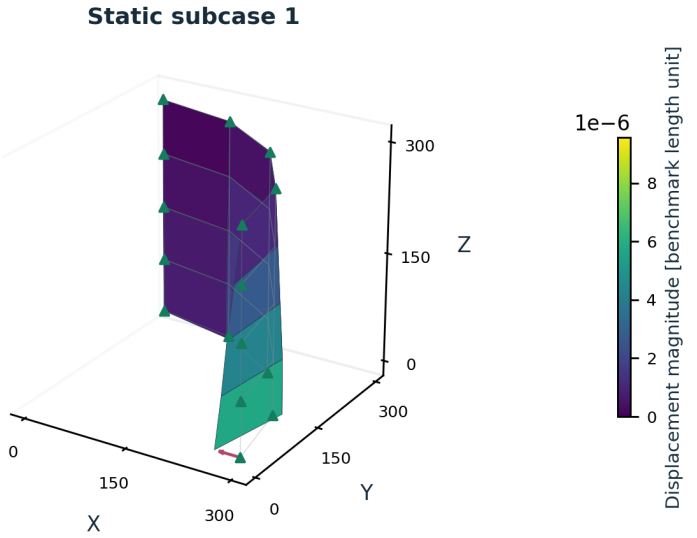
Pinched cylinder: coarse octant

MH_pinched_cylinder | SOL 101 | Published shell benchmark

Cylinder radius 300, full length 600, thickness 3; E=3e6, nu=0.3.

Geometry / mesh	25 nodes; 16 elements. Cylinder radius 300, full length 600, thickness 3; E=3e6, nu=0.3.
Material / section	E=3e+06, nu=0.3; shell thickness=3. Consistent benchmark units; displacements in deck length unit, forces in deck force unit, stress in force/length^2
Boundary conditions	Rigid end diaphragm and octant symmetry; quarter of unit radial point load at the symmetry intersection.
Applied load	Radial point force -0.25 in global X on the modeled octant symmetry corner.



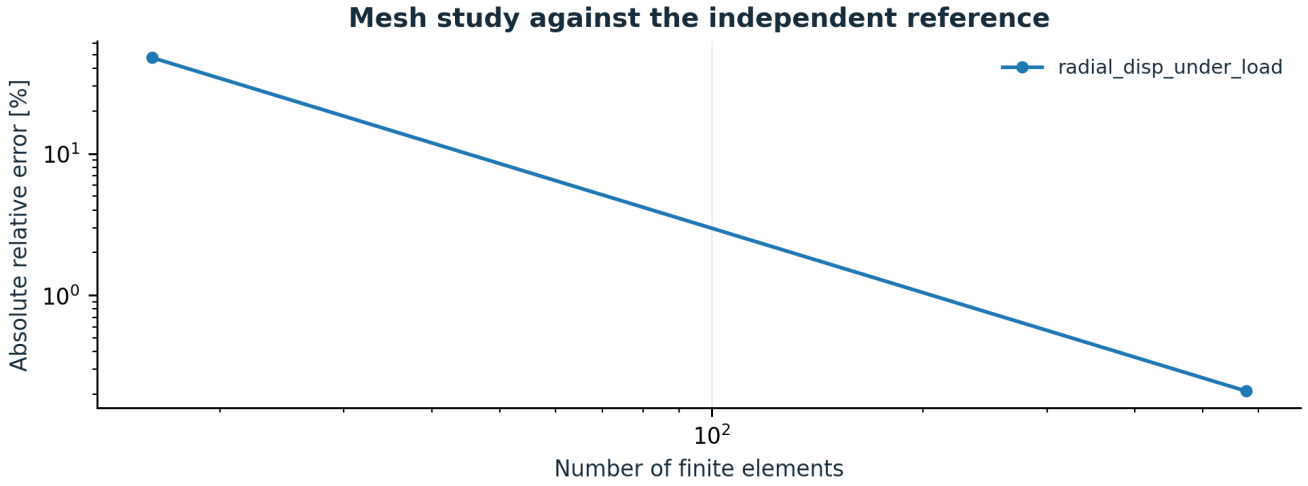
Shape amplified x3.76e+06; grey = original.

Recorded JFEM result; deformation is amplified, mode amplitudes/signs arbitrary. Green triangles mark constrained nodes; magenta arrows mark available point loads.

Recorded numerical checks

Quantity [unit]	JFEM	Reference	Error	Assessment
radial disp under load (magnitude) [benchmark length unit]	9.5636e-06	1.8248e-05	47.591%	FAIL (coarse)

Evidence, convergence and interpretation



The separately retained MH_pinched_cylinder_refined companion tests the same published target. Coarse-mesh error is displayed and is not relabeled as validation success.

Scope: Linear elasticity and small-displacement equilibrium or linearized bifurcation/modal analysis; this is computational verification, not a material-test validation.

Reference and reproduction

MacNeal, R.H.; Harder, R.L. (1985). A proposed standard set of problems to test finite element accuracy. Finite Elements in Analysis and Design 1, 3-20. https://doi.org/10.1016/0168-874X(85)90003-4

Ko, Y.; Lee, Y.; Lee, P.-S.; Bathe, K.-J. (2017). Performance of the MITC3+ and MITC4+ shell elements in widely-used benchmark problems. Computers & Structures 193, 187-206. https://doi.org/10.1016/j.compstruc.2017.08.003

Open the recorded .jfem file with Boxer: Project > Open JFEM results. For older eigenvalue files choose modal or buckling results when requested. The input deck and the recorded run receipt accompany the figures.

Run: 2026-10-19 | Solver revision: a70f997 | Full precision values: results.json / metrics.csv